NAME: ______________________ **PERIOD:** ________ **DATE:** ____________

One Line, Two Model Diagnostics

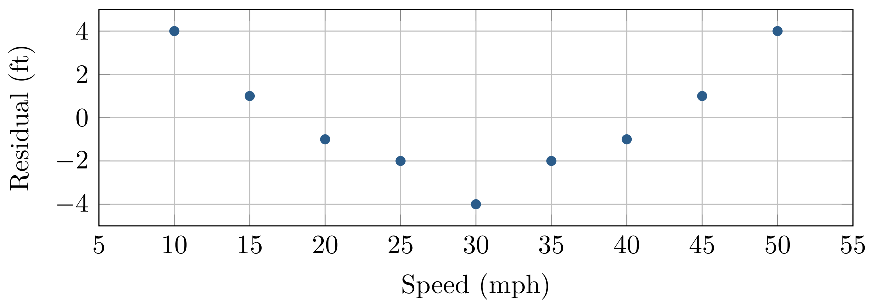
AP Statistics · Unit 5 · Topic 5.4 · B: 2027-02-11 · E: 2027-04-07

[STAR] TODAY'S PROBLEM

Suppose a road-safety lab records braking distance y at speeds x from 10 through 50 mph on one vehicle and dry surface. The line is $\hat{y} = -10 + 2.5x$, the correlation is $r = 0.997$, and the residual plot is shown. Actual distances at 10, 30, and 50 mph are 19, 61, and 119 feet.

Ezra: The large correlation and residuals on both sides of 0 justify using the line across the tested speeds.

June: I would use where the residuals occur and the 50-mph error to judge safety, then check any replacement model separately.



FIRST TAKE — before the video (5 min)

Which evidence should control the model judgment? Cite one numerical or graphical feature.

[BOOK] READING A RESIDUAL PLOT (keep on the page)

A residual is $y - \hat{y}$. A positive residual means the model underpredicted; a negative residual means it overpredicted. Plot residuals against x or $\hat{y}$. Random scatter around 0 supports a linear form. Curvature, trends, clusters, or changing spread show structure the line missed. Correlation measures the direction and strength of linear association, but even a large $|r|$ does not replace checking residual locations. Check a replacement model with its own residual plot.

Finish after the video:

DOK 1 **(a)** Read the residuals at 10, 30, and 50 mph and state whether the line underpredicts or overpredicts at each speed.

DOK 2 **(b)** Verify those residuals from the equation and actual distances. Describe the full residual plot, including its form and where the line tends to underpredict or overpredict.

DOK 3 **(c)** Evaluate both judgments by reconciling $r = 0.997$ with the residual locations. Assess what the large correlation and both residual signs do or do not establish, use the 50-mph error for a safety consequence, specify evidence required before trusting a replacement model, and bound the conclusion to the observed setting.

Frame: The correlation summarizes ____________________; the residual locations show ____________________.

Frame: At 50 mph the line predicts ________ while the actual value is ________, so for safety I would ____________________.

EXIT — turn this in

One thing the video changed about my first take: ____________________